Lab 9 Report: Validation of the Improved Build

MMAC Study Hub and Aura Screen

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At a glance. 5 new users did our two key tasks on the Lab 8 build, each on two versions of the Aura Screen report (A/B test). **All 10 runs of both tasks were completed without help** (Lab 7: 3 of 5). Average SUS rose from **63.5 to 83.5–85.5**. **6 of our 7 target checks were met**; the miss is T2 understanding (3/5 users fully correct, target 4/5). Of the 24 Lab 7 issues, **21 are fixed and 3 partly fixed**; every accessibility check now passes. **Version A** (short report) was 28 s faster on T1 than **Version B** (detailed report with charts), with the same understanding.

Key tasks (from Lab 5, unchanged since Lab 7):
T1 Aura Screen — create an account, set up your record, do a contactless scan, read your heart rate on the report.
T2 Study Hub — find all the data, explain one sample in your own words, request access to the full dataset.

1. User Validation

success criteria, methods, results, before vs after

1.1 Success criteria (set before testing)

We set four measurable targets for each task before testing. Comprehension, time and errors are checked per task, and SUS once for the whole system.

Target	How we measure it	Lab 7 (before)	Target (after)
1. Comprehension	3 questions after each task; user counts only if *all 3* are correct, unaided	Not measured; all 5 said they could not tell what results meant	$\geq$ 4 of 5 users, per task
2. Time	Seconds from reading the task to the success point	T1 218 s, T2 262 s	T1 $\leq$ 180 s, T2 $\leq$ 200 s
3. Errors	Wrong clicks, dead ends, going back (average per user)	T1 1.4, T2 1.8	$\leq$ 1.0 per task
4. SUS	Standard 10-item System Usability Scale	63.5	$\geq$ 68 (average)

1.2 Validation methods we chose

We used all three evaluation types from Lab 3.

Method	What we did	Why we chose it
User-based evaluation	Task-based testing with 5 new users (P6–P10, none from Lab 7), thinking aloud. We timed each task, counted errors and hints, asked 3 comprehension questions per task, then the SUS.	Measures targets 1–4 directly and shows whether users now *understand* the data, the main Lab 7 problem.
A/B test of the report	Each user did both tasks twice: once with report **Version A** (short, key results only) and once with **Version B** (longer, step-by-step explanation plus signal charts).	The Lab 7 fix for the report could go two ways (shorter or more explained); testing both shows which works better.
Usability inspection	Both of us walked T1 and T2 on the new build against Nielsen's 10 heuristics and marked every Lab 7 issue fixed, partly fixed or not fixed.	Finds layout and error-handling problems users rarely mention.
Accessibility evaluation	Repeated the Lab 7 checks, plus a DevTools contrast ratio, 200 % zoom and touch-target checks, and Lighthouse.	Checks the build works for low-vision and keyboard users.

Setup: laptop, Aura Screen in its phone view, Study Hub in the browser. Hint only after 60 s stuck; fail after 8 minutes. 13 October 2026.

1.3 Results from the new users

		T1 — Aura Screen			T2 — Study Hub				
User	Ver.	Time (s)	Errors	Understood	Time (s)	Errors	Understood	SUS	Outcome
P6	A	142	2	3/3	150	0	3/3	85.0	both alone
	B	158	0	3/3	128	0	3/3	77.5	both alone
P7	A	118	2	2/3	120	1	2/3	70.0	both alone
	B	134	0	3/3	70	1	2/3	92.5	both alone
P8	A	96	0	3/3	105	0	3/3	95.0	both alone
	B	171	1	2/3	94	1	2/3	72.5	both alone
P9	A	90	0	3/3	122	0	3/3	80.0	both alone
	B	122	0	3/3	102	0	3/3	92.5	both alone
P10	A	121	0	3/3	95	1	2/3	87.5	both alone
	B	121	0	3/3	92	0	3/3	92.5	both alone
Mean	A	**113**	**0.8**	**4/5**	**118**	**0.4**	**3/5**	**83.5**	5/5 alone
	B	**141**	**0.2**	**4/5**	**97**	**0.4**	**3/5**	**85.5**	5/5 alone

"Understood" = comprehension questions correct; the mean row counts users with all 3 correct. No answer was wrong; the 6 misses were all "partly correct". SUS is recomputed from each user's ticked answers (three hand-totals on the forms were off by 2.5–7.5 points).

What users said. No one failed or needed a hint. The camera was blocked in 6 of 10 runs, and **every time the on-screen message let the user fix it** (Lab 7: P1 was stuck until told). Users rated the explanations well: "Aura Screen explained my results in words I understood" averaged **4.8/5** in both versions; "The website explained each sample dataset" averaged 4.2 (A) and 3.8 (B). Comments that point to remaining problems: no sign that the scan has started (P6, wants a progress bar); hard-to-see fonts (P7); access form has too many fields (P10); "EMG/ECG sample" answers were sometimes vague (T2 partly correct for P7, P8, P10).

1.4 Before vs after (Lab 7 → Lab 9)

Measure	Lab 7	Lab 9, Ver. A	Lab 9, Ver. B	Change
Completed without help, T1 / T2	3/5 / 3/5	5/5 / 5/5	5/5 / 5/5	+2 users each
Mean time, T1	218 s*	113 s	141 s	35–48 % faster
Mean time, T2	262 s*	118 s	97 s	55–63 % faster
Mean errors, T1 / T2	1.4 / 1.8*	0.8 / 0.4	0.2 / 0.4	fewer
Users who understood the results, T1 / T2	0/5 (all confused)	4/5 / 3/5	4/5 / 3/5	main fix
Mean SUS	63.5 (below average)	83.5	85.5	+20 to +22

*Lab 7 times and errors were recorded as estimates, so treat those changes as indicative; completion and SUS come from the filled forms in both labs.

1.5 A/B test of the session report

	Version A — short report	**Version B** — detailed report with charts
T1 mean time	**113 s**	141 s (5 of 5 users as slow or slower)
T1 understood (all 3)	4/5	4/5
T1 errors	0.8	0.2
What users said	"The report — too little detail" (P9); "more explanation"	"Scrolled a little longer for more data" (P6); "lots of scrolling before the chart ends"; "charts too much — remove charts" (P8); "show the explanation when I tap" (P9)

Result: both versions were understood equally well, but B cost about 28 s more on T1 and drew complaints about scrolling and charts, while A was called too thin. **Decision:** keep A's short report as the default and put B's step-by-step explanation and charts behind a "Show how we got this" tap. Each user tried both versions; the order was not recorded, so practice may favour the second run (T2, identical in both, was 21 s faster in B).

Screens of both versions: Figure 1, page 6. What each comment changes, incl. the reasoning model: Section 4.2, page 8.

2. Inspection & Accessibility *status of Lab 7 issues, accessibility results*

2.1 Status of the Lab 7 issues

Both of us re-checked every issue from Lab 7 (17 from the heuristic inspection, 7 found only by users) on the Lab 8 build. Result: **21 fixed, 3 partly fixed, 0 not fixed**. Figure numbers refer to the Lab 8 before/after report.

ID	Lab 7 problem	Sev	What we changed (Lab 8 Fig)	Status
Aura Screen				
A1	Phone/kiosk view only via dev tools	3	Phone / Kiosk / Full-screen toggle (13)	**Fixed**
A2	Scan Hub scrolls in phone view	3	No-scroll phone screens	**Fixed**
A3	Third heart-rate tile cut off	3	Shorter, fixed report (13); *Version B still needs long scrolling*	**Partly**
A4	Navigation out of thumb reach	2	Bottom tabs: Scan, Report, Timeline (13)	**Fixed**
A5	13 px inputs make iOS zoom	2	16 px inputs	**Fixed**
A6	Hover stays after a tap	1	Hover effect only on mouse devices	**Fixed**
A7	Kiosk screens leave empty space	2	Kiosk-first layout	**Fixed**
Study Hub website				
W1	Dashboard links lead nowhere	4	Dashboards built into the site; **EMG and rPPG dashboards added**; dead demo removed (4)	**Fixed**
W2	Dashboards are recording tools	3	View-only dashboards, 6 widgets each (9, 10)	**Fixed**
W3	Dashboards look unlike the site	3	One shared layout for all four dashboards (9, 10)	**Fixed**
W4	Site fails to start	3	One-command launcher + README	**Fixed**
W5	Placeholder text	2	Placeholder card removed (4)	**Fixed**
W6	Open downloads, no terms of use	2	Request access + data use agreement (12)	**Fixed**
W7	Charts work with a mouse only	2	Widgets reachable and movable with the keyboard	**Fixed**
W8	Draggable widgets, no hint	1	Grip icon + “Drag widgets by their title” (9)	**Fixed**
W9	Request needs an email app	2	Address shown to copy if no email app opens	**Fixed**
W10	Thermal values are illustrative	1	“Public sample” badge; sample explained (7)	**Fixed**
Found by users in Lab 7				
U1	Report numbers not explained	3	Plain verdict + steps with the normal range (13)	**Fixed**
U2	Sample data has no explanation	3	*About this data*, a caption on every chart, full details (7–12)	**Fixed**
U3	Camera blocked, no message	3	Camera errors explained on screen (worked in 6/6 blocked runs)	**Fixed**
U4	Hard to find the data	2	*Explore the data* button, *Data* menu with descriptions, links, footer (1–3, 6)	**Fixed**
U5	Access form long and dense	2	3 short steps + “In short” summary (12)	**Fixed**
U6	Google is the only sign-in	1	Email + password added; phone sign-in from the plan not built	**Partly**
U7	Unclear how the two products relate	1	*Built with the data* section on the website (4); not yet on Aura Screen	**Partly**

New problems: the inspection found none. Users raised four small ones, listed under *Still pending* (Section 3).

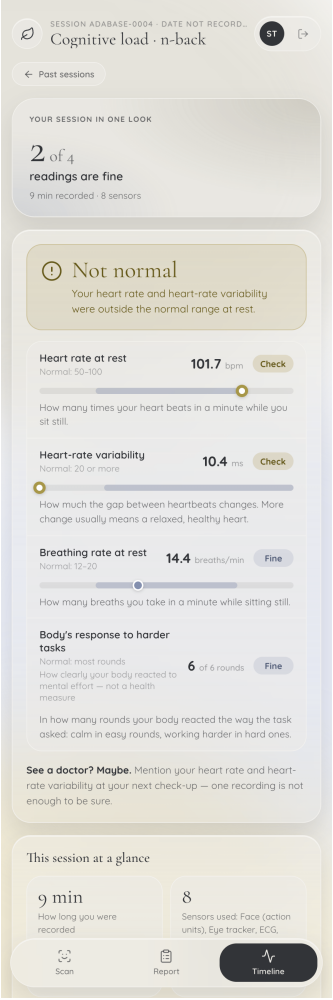
2.2 Accessibility check

Check	How we checked it	Lab 7	Lab 9
Text contrast	Faded grey text; DevTools contrast ratio ≥ 4.5:1. **Lowest found: 4.59:1**	Fail	**Pass**
Chart labels	Axis and phase labels readable at a glance (larger, brighter — Lab 8 Fig 11)	Fail	**Pass**
Text size	Captions, labels and badges big enough	Partly	**Pass**
Keyboard	Whole site with keyboard only: menu, dashboards, request form	Partly	**Pass**
Labels	Images and icon buttons have names (theme toggle, close)	Partly	**Pass**
Zoom / touch	200 % zoom loses nothing; Aura Screen buttons ≥ 44 px	new	**Pass**
Lighthouse	Chrome DevTools audit, Mobile; Aura Screen home: **96/100** (Fig. 3c)	new	**Pass**
Colour blindness	Chrome DevTools, *Emulate vision deficiencies* (all four types): Fine/Check written in words, range-bar dot shows position, never colour alone	new	**Pass**
Language	English / Hindi (हिं) switch on Aura Screen: headings, buttons, labels and readings translated, e.g. “बिना छुए स्कैन शुरू करें” (start contactless scan)	new	**Pass**

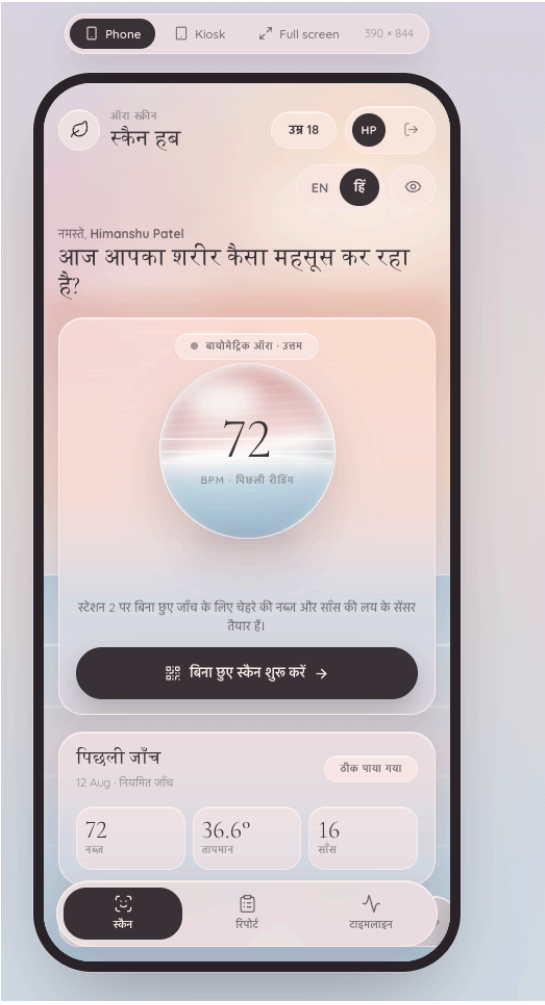
All three Lab 7 “Partly” checks and both “Fail” checks now pass, and so do the four new checks (Figure 3, page 4).

2.3 Figure 3 — accessibility features and the Lighthouse audit

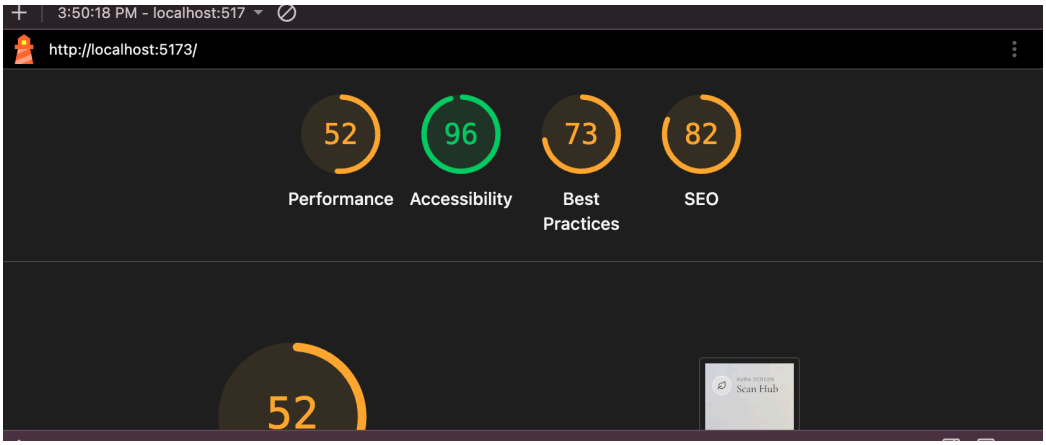
Both screens are Aura Screen in the phone view used for testing.



(a) Colour blindness. The session report checked with Chrome DevTools *Emulate vision deficiencies*. Status is still clear without colour: the words *Check* and *Fine*, the warning icon, and where the dot sits on each range bar.



(b) Hindi. The scan home after tapping हिं (top right, next to EN). The greeting, the scan button (बिना छुए स्कैन शुरू करें), the last check's readings and the bottom tabs are all in Hindi.



(c) Lighthouse (Chrome DevTools, Mobile, run on `localhost:5173`, Aura Screen home). **Accessibility 96/100.** The other scores (Performance 52, Best Practices 73, SEO 82) are outside this lab's accessibility check; Performance is low partly because this is the unoptimised development server.

3. Validation Summary *targets met, still pending, plan to finish*

3.1 Which targets were met

Target	Goal	Version A	Version B	Result
1. Comprehension, T1	$\geq 4/5$	4/5	4/5	**Met**
1. Comprehension, T2	$\geq 4/5$	3/5	3/5	**Not met**
2. Time, T1 / T2	≤ 180 / 200 s	113 / 118 s	141 / 97 s	**Met**
3. Errors, T1 / T2	≤ 1.0	0.8 / 0.4	0.2 / 0.4	**Met**
4. SUS	≥ 68	83.5	85.5	**Met**

6 of 7 checks met in both versions. T2 comprehension missed by one user: 3 users got the T2 questions only partly right (which four signals the study records, and what one sample shows).

3.2 What is still pending

- **T2 comprehension (target not met)**: users cannot always name all four signals or say what a sample shows in one sentence.
- **Report length (A/B result)**: Version B is too long, Version A too thin; this is also why A3 is only partly fixed.
- **Partly fixed issues**: U6 (no phone sign-in yet), U7 (link between the products shown only on the website).
- **New user comments**: no progress indicator once the scan starts (P6); hard-to-see fonts (P7); too many fields on the access form (P10).
- **Evidence gaps**: Lighthouse scores passed but were not written down; all users were students, tested on a laptop in phone view; A/B order not recorded.

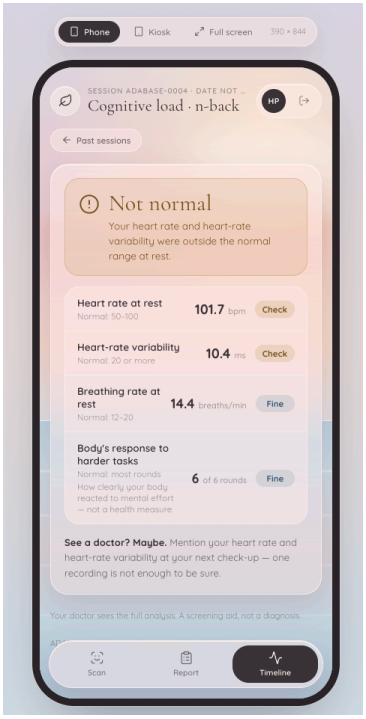
3.3 Plan to complete the project (final presentation, ~31 Oct)

#	Action	Who	By
1	Merge A and B: short report by default, step-by-step explanation and charts behind "Show how we got this"	Himanshu	20 Oct
2	Fix T2 understanding: a one-sentence "what this sample shows" line at the top of each dashboard; name all four signals on the home page	Hans	22 Oct
3	Add a scan progress indicator; increase small font sizes	Himanshu	22 Oct
4	Cut access-form fields; finish U6 (phone sign-in) and U7 (link to the Study Hub from Aura Screen)	Hans, Himanshu	24 Oct
5	Record Lighthouse scores; quick re-test with 3 users including a non-student on a real phone, alternating A/B order	Both	27 Oct
6	Improve the reasoning model (Section 4.1, steps 1–2): numbers only from the evidence record; show a plain reason when it abstains or withholds	Both	29 Oct
7	Final presentation and report	Both	31 Oct

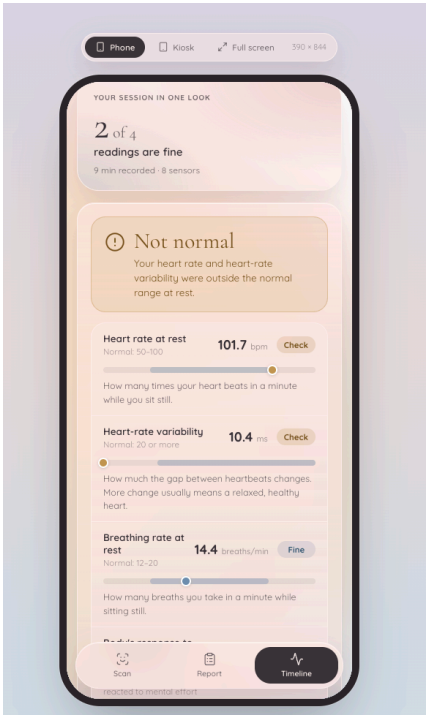
Next round: Section 4 (pages 7–9) rates the remaining problems by Nielsen heuristic and severity, focusing on the reasoning model, with a plan to improve it; 4.3 shows the model's architecture.

1.5 (continued) Figure 1 — the two report versions tested

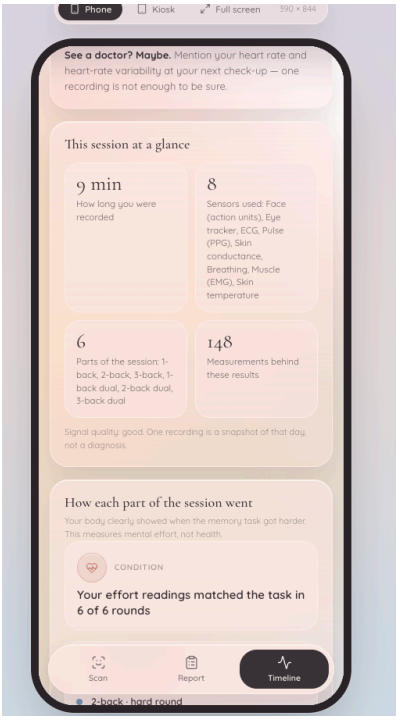
Both screens show the same session (ADABase-0004) in the phone view (390×844) used in testing. Version A ends after the readings list and a one-line advice; Version B adds a range bar and a meaning under every reading, then a *session at a glance* card and *how each part of the session went*, so it needs several screens of scrolling.



(a) Version A — short report. Verdict, four readings with Fine/Check, one line of advice.



(b) Version B — top. "2 of 4 readings are fine", then each reading with a range bar and meaning.



(c) Version B — scrolled down. Session-at-a-glance card and the per-round results; charts follow below.

	Version A (a)	**Version B** (b, c)
Fits on one screen	Yes, apart from the footer line under the tabs	No — about 4 screens before the charts end
Explains each reading	Normal range only	Range bar + one-line meaning
T1 time / understood	113 s / 4 of 5	141 s / 4 of 5
Main complaint	"Too little detail" (P9)	Scrolling and charts (P6, P8)

What the screens show that users did not say

- The verdict *Not normal* is driven by a heart rate of 101.7 bpm against a limit of 100 — a value just over the line is shown with the same weight as a large deviation, and with no sign of how sure the system is.
- *Body's response to harder tasks* is the reasoning model's output (cognitive load), but it sits in the same list as the vital signs, with the same Fine/Check badge, while its own caption says it is "not a health measure".
- The header shows an internal session ID and a cut-off "Date not …"; in (a) and (b) the fixed bottom tabs cover the last lines of content.

4. Heuristic Issues for the Next Iteration *Nielsen heuristic and severity*

ID	Heuristic	Problem	Sev	What we will change
R1	H2 Match with the real world; H1 Visibility	**The report's explanation is hard to follow and trust.** 39 % of sessions get no explanation and the screen does not say why. 17 % of first drafts make up a number. All 8 sensors are listed as reasons, though each alone is close to chance (0.44–0.54). The doctor page talks in log-odds.	3	Numbers are filled in from the session data, never written by the language model. If there is no explanation, say why in one line and give a next step. Show only the sensors that really changed the result, as a percentage. Each sentence links to the data behind it.
R2	H1 Visibility of system status	**No progress shown in either task.** T1: nothing shows the scan has started or how long is left (P6). T2: after the access request is sent, the user is not told what happens next.	2	T1: a progress ring with seconds left. T2: a step bar on the form (1 of 3) and a request status: sent, being reviewed, approved.
R3	H3 User control and freedom	**Sharing with the doctor is not clear enough.** The patient cannot easily pick which sessions to share, see whether the doctor has opened them, or stop sharing. The doctor cannot reply or mark an explanation as wrong.	2	A "Share with my doctor" button on each report: choose sessions, see "Viewed on 12 Oct", stop sharing any time. The doctor can add a note and flag a wrong explanation, which we keep to improve the model.
R4	H10 Help; H4 Consistency	**The model does not say how sure it is or where it can be trusted.** 101.7 bpm against a limit of 100 looks as certain as a big miss. It was trained only on one lab memory task (score 0.59 on a different task). The mental-effort result has the same Fine/Check badge as heart rate (Fig. 1a).	3	Show how sure the model is, with a "borderline" label near limits. Add a short "What this model was trained on" note. Move mental effort to its own card with its own scale, never a health badge.
R5	H1 Visibility; H3 Control	**Users cannot see who used their data.** Nothing shows whether a doctor, the model or a researcher used a session, or whether it was used to train the model.	3	A "Who used my data" list on the patient side (doctor viewed, model analysed, shared on). A separate switch for "Use my data to improve the model".

Severity: 1 cosmetic, 2 minor, 3 major, 4 catastrophic. Three major and two minor issues; R1 and R4 are in the reasoning model.

4.1 Plan to make the reasoning model better (steps 1–2 by 29 Oct, row 6 of the plan in 3.3; the rest after the final presentation)

#	Change	How we will know (now → target)	Fixes
1	The language model writes the sentences; the numbers come from the session data	Made-up numbers on first try 17 % → under 2 %; held back 15 % → under 5 %	R1
2	A plain reason and a next step whenever there is no explanation	Sessions with an explanation or a reason: 61 % → 100 %	R1
3	List only the sensors that changed the result; check by removing the top one	Share of explanations where removing the top sensor changes the result	R1
4	Show how sure the model is; add the "trained on" note; own card for mental effort	Confidence shown on every report; no mental-effort result with a health badge	R4
5	Doctor's "wrong explanation" flags saved as examples to retrain on	Flags logged per model version	R3
6	Re-test with 3–5 users, including a doctor: can they say in their own words why the model decided?	Answer matches the session data for at least 4 of 5	all

4.2 From user results to changes — screen, reasoning model, and what deployment adds

Each row starts from what a Lab 9 user said or did (Section 1.3, Appendix A). The last column says what we would gain if Aura Screen were deployed to real patients rather than tested on a laptop in phone view.

User(s)	What they said / did	Fix in the app	Reasoning-model improvement	If deployed
P6	No sign the scan had started; wants a progress bar	Progress ring with seconds left (plan 3)	Show live signal quality and drop bad windows before analysis, so fewer sessions end in an abstain	Real phone cameras and lighting vary; logged signal quality shows which conditions cause abstains
P7	Fonts hard to see	Raise small text, keep contrast $\geq 4.5{:}1$ (plan 3)	Show the explanation at the same readable size; check every explanation with a readability score	Respects the phone's own text-size setting; reaches older and low-vision users we did not test
P10	Access form has too many fields	Cut to essential fields (plan 4)	Record consent once; the same switch decides whether a patient's data may train the model	A visible data-use log shows who (doctor, model, researcher) used each person's data
P9	Version A gives too little detail	"Show how we got this" tap (plan 1)	That tap opens the model's verified explanation: numbers only from the evidence record (plan 6)	Count how often people tap it, and which explanations they open, to learn what needs explaining
P6, P8	Version B: too much scrolling, too many charts	Charts behind the tap (plan 1)	List only sensors that really moved the result (R1), so the explanation is short	Scroll-depth data from many users replaces our 5-person judgement
P7, P8, P10	T2 answers only partly right (signals, what a sample shows)	One-sentence "what this sample shows" line (plan 2)	Same verified explainer can write that sentence for each sample, checked against the data	Study Hub visitors' answers can be sampled continuously, not once
6 of 10 runs	Camera blocked; the message let every user fix it	Keep the message	When the signal is too poor, say so in plain words and suggest a rescan (R1)	Error reports show how often scans fail in homes or clinics

What deployment would add to the reasoning model

- **More and broader data:** ADABase has 30 lab subjects doing one memory task; real sessions from non-students on real phones would let us recalibrate the model and measure how it holds up outside the n-back setting (cross-task is now 0.59).
- **Labelled mistakes:** "This explanation is wrong" and "redo this reading" flags (R3) become training labels for the explainer and the verifier.
- **Live quality numbers:** invented-number, withheld and abstain rates (now 17 %, 15 %, 24 %) can be tracked per model version, so each change is checked on real use.
- **Limits first:** before deployment the model needs validation on real users, clear consent and a data-use log, and the "screening aid, not a diagnosis" message must stay on every report.

4.3 The reasoning model we are building for the next version

One rule runs the whole design: **a machine-learning model makes the decision; a small language model only explains it.** It already runs offline on the ADABase data and fills the doctor's *What the model says* page for our demo sessions. Making it part of every live scan, with the changes in 4.1, is the work for the next version.

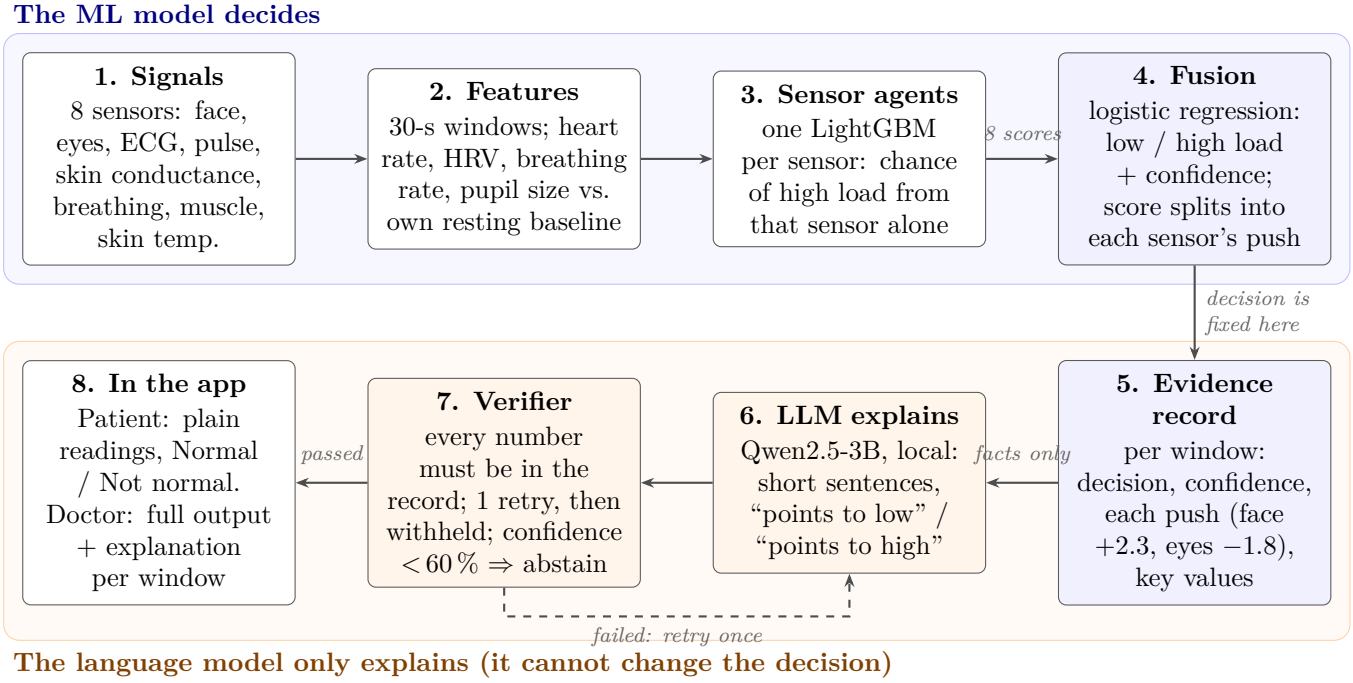


Figure 2: architecture of the reasoning model. Blue: decision path. Orange: explanation path.

Why it is built this way	What it gives us
Decision and explanation are separate	The language model cannot talk its way to a different answer; it only describes the fusion's result
Score splits into pushes per sensor	Every "points to high load" line has an exact number behind it, so the explanation is faithful by construction
Verifier + abstain	No made-up numbers reach a user; a weak case says "not sure" instead of guessing
Runs locally (Ollama)	Health data never leaves the machine to reach an outside AI service

Results so far (leave-one-subject-out, 30 people): fused AUROC **0.87**, against 0.44–0.54 for any single sensor, so combining sensors is what makes it work. Every explanation shown is grounded in numbers from the record. Still to fix (Section 4, R1 and R4): 17 % of first drafts make up a number, 15 % are withheld, 24 % abstain, and on a different task the score drops to 0.59.

4.4 Future plan: tracing who uses the Study Hub dataset

Our data use agreement says the dataset must not be passed to any other institution. To make that checkable, each approved researcher will get **their own copy with an invisible fingerprint**:

1. **Fingerprint the signals.** A few hundred chosen samples are shifted by far less than sensor noise (e.g. 0.001 µS on skin conductance). The pattern of shifts encodes the recipient's ID; it does not change any research result.
2. **Hidden ID in the file.** The recipient ID also goes into file metadata and column order, as a quick first check (easy to strip, so not relied on alone).
3. **Download register.** Every approved download is logged: who, when, which version, which fingerprint.
4. **Check a leaked copy.** If a copy appears elsewhere (another institution, GitHub, Kaggle), we read its fingerprint and match it to the register to find whose copy it was.
5. **State it in the agreement.** The agreement says each copy is fingerprinted and forbids sharing or uploading the data to AI services; knowing this is itself a deterrent.

Limits: it traces *which copy* leaked, not every use; proving AI training needs canary records and access to the model. Only for our own recordings, not ADABase.

Appendix A: Filled participant forms

Version A then B for each user; blank second pages removed

P6 — Version A

Lab 9 Validation — Participant P___

Version A

one per participant

Name: Ravi Entry no.: 2024aib1012 Date: 13/10/2024

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☑ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 142 s Errors: 2 ☑ Completed ☐ With help ☐ Failed

Camera: ☑ worked ☐ blocked, the message helped ☐ blocked, stuck

Notes / hints: Went to report, then to timeline

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 150 s Errors: 0 ☑ Completed ☐ With help ☐ Failed Found data via: ____

Notes / hints: ____

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows 79 bpm)		☑	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)		☑	☐	☐
T1.3	Should you do anything next? (as the report says)		☑	☐	☐
T2.1	Which signals does the study record? (all four)		☑	☐	☐
T2.2	What does one sample show? (the signal + one finding)		☑	☐	☐
T2.3	Can you download the full data now? (no: request + agreement)		☑	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☑	☐
2	I found the system unnecessarily complex.	☐	☑	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☑
4	I think that I would need the support of a technical person to use this system.	☑	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☑	☐
6	I thought there was too much inconsistency in this system.	☐	☑	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☑
8	I found the system very cumbersome to use.	☑	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☑	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☑	☐	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☐	☑
E2	The website explained what each sample dataset shows.	☐	☐	☐	☑	☐

Debrief

Most confusing moment: Waiting for scan, not sure it has started

What would you change first? A progress bar

Facilitator: SUS score 85 G3 met: T1 ☐ T2 ☐

Version B

Lab 9 Validation — Participant P___

one per participant

Name: Ravi Entry no.: 2024aib1012 Date: _____

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☐ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 158 s Errors: 0 ☑ Completed ☐ With help ☐ Failed

Camera: ☑ worked ☐ blocked, the message helped ☐ blocked, stuck

Notes / hints: Scrolled a little longer for more data

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 128 s Errors: 0 ☑ Completed ☐ With help ☐ Failed Found data via: Explore button

Notes / hints: _____

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows 79.9 bpm)		☑	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)		☑	☐	☐
T1.3	Should you do anything next? (as the report says)		☑	☐	☐
T2.1	Which signals does the study record? (all four)		☑	☐	☐
T2.2	What does one sample show? (the signal + one finding)		☑	☐	☐
T2.3	Can you download the full data now? (no: request + agreement)		☑	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☑	☐
2	I found the system unnecessarily complex.	☐	☑	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☑	☐
4	I think that I would need the support of a technical person to use this system.	☑	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☑	☐
6	I thought there was too much inconsistency in this system.	☐	☑	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☑	☐
8	I found the system very cumbersome to use.	☐	☑	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☑	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☑	☐	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☐	☑
E2	The website explained what each sample dataset shows.	☐	☑	☐	☐	☐

Debrief

Most confusing moment: Lots of scrolling before chart ends

What would you change first? Better navigation

Facilitator: SUS score 77.5 G3 met: T1 ☐ T2 ☐

P7 — Version A

Lab 9 Validation — Participant P2

one per participant

Name: Aryan Entry no.: 2024csb1102 Date: _____

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☐ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 118 s Errors: 2 ☑ Completed ☐ With help ☐ Failed

Camera: ☐ worked ☑ blocked, the message helped ☐ blocked, stuck

Notes / hints: _____

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 120 s Errors: 1 ☑ Completed ☐ With help ☐ Failed Found data via: Data Menu

Notes / hints: Opened about page first

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows _____ bpm)		☑	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)		☐	☑	☐
T1.3	Should you do anything next? (as the report says)		☑	☐	☐
T2.1	Which signals does the study record? (all four)		☐	☑	☐
T2.2	What does one sample show? (the signal + one finding)		☑	☐	☐
T2.3	Can you download the full data now? (no: request + agreement)		☑	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☑	☐	☐
2	I found the system unnecessarily complex.	☐	☑	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☑	☐
4	I think that I would need the support of a technical person to use this system.	☐	☑	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☑	☐
6	I thought there was too much inconsistency in this system.	☐	☑	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☑	☐
8	I found the system very cumbersome to use.	☐	☑	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☑	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☐	☑	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☑	☐
E2	The website explained what each sample dataset shows.	☐	☐	☐	☑	☐

Debrief

Most confusing moment: The camera

What would you change first? Pop-more explainable

Facilitator: SUS score 76 G3 met: T1 ☑ T2 ☑

P7 — Version B

Lab 9 Validation — Participant P___

one per participant

Name: Aryan Entry no.: 2022csb1112 Date: _____

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☐ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 1345 s Errors: 0 ☐ Completed ☐ With help ☐ Failed

Camera: ☐ worked ☐ blocked, the message helped ☐ blocked, stuck

Notes / hints: _____

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 70 s Errors: 1 ☐ Completed ☐ With help ☐ Failed Found data via: Data Menu

Notes / hints: Open Eng dash board before request my data

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows _____ bpm)		☐	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)		☐	☐	☐
T1.3	Should you do anything next? (as the report says)		☐	☐	☐
T2.1	Which signals does the study record? (all four)		☐	☐	☐
T2.2	What does one sample show? (the signal + one finding)		☐	☐	☐
T2.3	Can you download the full data now? (no: request + agreement)		☐	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
2	I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☐
4	I think that I would need the support of a technical person to use this system.	☐	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
8	I found the system very cumbersome to use.	☐	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☐	☐
E2	The website explained what each sample dataset shows.	☐	☐	☐	☐	☐

Debrief

Most confusing moment: Fonts — visibility

What would you change first? Better Fonts

Facilitator: SUS score 92.5 G3 met: T1 ☐ T2 ☐

P8 — Version A

Lab 9 Validation — Participant P___

one per participant

Name: Yash Entry no.: 2024eeb1235 Date: _____

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☐ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 96 s Errors: 0 ☑ Completed ☐ With help ☐ Failed

Camera: ☐ worked ☑ blocked, the message helped ☐ blocked, stuck

Notes / hints: _____

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 105 s Errors: 0 ☑ Completed ☐ With help ☐ Failed Found data via: _____

Notes / hints: _____

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows _____ bpm)		☑	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)		☑	☐	☐
T1.3	Should you do anything next? (as the report says)		☑	☐	☐
T2.1	Which signals does the study record? (all four)		☑	☐	☐
T2.2	What does one sample show? (the signal + one finding)		☑	☐	☐
T2.3	Can you download the full data now? (no: request + agreement)		☑	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☑	☐
2	I found the system unnecessarily complex.	☑	☐	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☑
4	I think that I would need the support of a technical person to use this system.	☑	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☑
6	I thought there was too much inconsistency in this system.	☑	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☑
8	I found the system very cumbersome to use.	☐	☑	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☑
10	I needed to learn a lot of things before I could get going with this system.	☑	☐	☐	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☐	☑
E2	The website explained what each sample dataset shows.	☐	☐	☐	☐	☑

Debrief

Most confusing moment: No thing

What would you change first? _____

Facilitator: SUS score 95 G3 met: T1 ☐ T2 ☐

P8 — Version B

Lab 9 Validation — Participant P___

one per participant

Name: Yash Entry no.: 2024eb1235 Date: _____

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☐ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 171 s Errors: 1 ☑ Completed ☐ With help ☐ Failed

Camera: ☐ worked ☑ blocked, the message helped ☐ blocked, stuck

Notes / hints: followed message and solved, cards too much

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 24 s Errors: 1 ☑ Completed ☐ With help ☐ Failed Found data via: _____

Notes / hints: missed explore button

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows _____ bpm)		☑	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)		☐	☑	☐
T1.3	Should you do anything next? (as the report says)		☑	☐	☐
T2.1	Which signals does the study record? (all four)		☑	☐	☐
T2.2	What does one sample show? (the signal + one finding)		☐	☑	☐
T2.3	Can you download the full data now? (no: request + agreement)		☑	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☑	☐	☐
2	I found the system unnecessarily complex.	☐	☑	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☑	☐
4	I think that I would need the support of a technical person to use this system.	☐	☑	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☑	☐
6	I thought there was too much inconsistency in this system.	☐	☑	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☑	☐
8	I found the system very cumbersome to use.	☐	☑	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☑	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☑	☐	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☑	☐
E2	The website explained what each sample dataset shows.	☐	☐	☐	☑	☐

Debrief

Most confusing moment: Charts

What would you change first? Remove Visuals

Facilitator: SUS score 72.5 G3 met: T1 ☐ T2 ☐

P9 — Version A

Lab 9 Validation — Participant P___

one per participant

Name: Manan Entry no.: 2024csb1128 Date: _____

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☐ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 90 s Errors: 0 ☑ Completed ☐ With help ☐ Failed

Camera: ☑ worked ☐ blocked, the message helped ☐ blocked, stuck

Notes / hints: The report — too less detail

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 122 s Errors: 0 ☑ Completed ☐ With help ☐ Failed Found data via: _____

Notes / hints: _____

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows _____ bpm)		☑	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)		☑	☐	☐
T1.3	Should you do anything next? (as the report says)		☑	☐	☐
T2.1	Which signals does the study record? (all four)		☑	☐	☐
T2.2	What does one sample show? (the signal + one finding)		☑	☐	☐
T2.3	Can you download the full data now? (no: request + agreement)		☑	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☑	☐
2	I found the system unnecessarily complex.	☐	☑	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☑	☐
4	I think that I would need the support of a technical person to use this system.	☐	☑	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☑
6	I thought there was too much inconsistency in this system.	☐	☐	☑	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☑	☐
8	I found the system very cumbersome to use.	☐	☑	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☑
10	I needed to learn a lot of things before I could get going with this system.	☑	☐	☐	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☐	☑
E2	The website explained what each sample dataset shows.	☐	☐	☐	☑	☐

Debrief

Most confusing moment: Less data — report

What would you change first? more explanation

Facilitator: SUS score 82 G3 met: T1 ☐ T2 ☐

P9 — Version B

Lab 9 Validation — Participant P___

one per participant

Name: Mahan Entry no.: 2024csb1128 Date: _____

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☐ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 122 s Errors: 0 ☑ Completed ☐ With help ☐ Failed

Camera: ☐ worked ☑ blocked, the message helped ☐ blocked, stuck

Notes / hints: _____

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 102 s Errors: 0 ☑ Completed ☐ With help ☐ Failed Found data via: _____

Notes / hints: _____

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows _____ bpm)		☑	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)		☑	☐	☐
T1.3	Should you do anything next? (as the report says)		☑	☐	☐
T2.1	Which signals does the study record? (all four)		☑	☐	☐
T2.2	What does one sample show? (the signal + one finding)		☑	☐	☐
T2.3	Can you download the full data now? (no: request + agreement)		☑	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☑	☐
2	I found the system unnecessarily complex.	☑	☐	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☑
4	I think that I would need the support of a technical person to use this system.	☑	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☑	☐
6	I thought there was too much inconsistency in this system.	☑	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☑
8	I found the system very cumbersome to use.	☑	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☑
10	I needed to learn a lot of things before I could get going with this system.	☐	☑	☐	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☐	☑
E2	The website explained what each sample dataset shows.	☐	☐	☐	☐	☑

Debrief

Most confusing moment: Nothing

What would you change first? show explanation when I tap

Facilitator: SUS score 92.5 G3 met: T1 ☑ T2 ☐

P10 — Version A

Lab 9 Validation — Participant P___

one per participant

Name: Shlok Entry no.: 2024aib1018 Date: _____

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☐ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 121 s Errors: 0 ☑ Completed ☐ With help ☐ Failed

Camera: ☑ worked ☐ blocked, the message helped ☐ blocked, stuck

Notes / hints: _____

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 95 s Errors: 1 ☑ Completed ☐ With help ☐ Failed Found data via: _____

Notes / hints: Missed the section

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows _____ bpm)	112	☑	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)	72-96	☑	☐	☐
T1.3	Should you do anything next? (as the report says)	Doctor will review	☑	☐	☐
T2.1	Which signals does the study record? (all four)	4	☑	☐	☐
T2.2	What does one sample show? (the signal + one finding)	Ans	☐	☐	☐
T2.3	Can you download the full data now? (no: request + agreement)	No, request access	☑	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☑	☐
2	I found the system unnecessarily complex.	☐	☑	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☑	☐
4	I think that I would need the support of a technical person to use this system.	☑	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☑	☐
6	I thought there was too much inconsistency in this system.	☑	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☑
8	I found the system very cumbersome to use.	☐	☑	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☑
10	I needed to learn a lot of things before I could get going with this system.	☑	☐	☐	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☐	☑
E2	The website explained what each sample dataset shows.	☐	☐	☐	☑	☐

Debrief

Most confusing moment: The access form

What would you change first? Fewer fields

Facilitator: SUS score 80 G3 met: T1 ☐ T2 ☐

P10 — Version B

Lab 9 Validation — Participant P___

one per participant

Name: Shlok Entry no.: 2024aib1014 Date: _____

☐ Not in the Lab 7 test ☐ I consent Aura Screen on: ☐ real phone ☐ phone view on laptop

T1 — Aura Screen

"Create an account, finish setting up your record, start a contactless scan, then open your session report and tell us your heart rate."

Time: 121 s Errors: 0 ☑ Completed ☐ With help ☐ Failed

Camera: ☐ worked ☑ blocked, the message helped ☐ blocked, stuck

Notes / hints: _____

T2 — Study Hub website

"Find all the data available on the site. Open the sample data and tell us in your own words what it shows. Then request access to the full dataset."

Time: 92 s Errors: 0 ☑ Completed ☐ With help ☐ Failed Found data via: _____

Notes / hints: _____

Understanding ✓ correct ∼ partly × wrong / don't know

	Question (what counts as correct)	Answer	✓	∼	×
T1.1	What was your heart rate? (report shows _____ bpm)	79.9	☑	☐	☐
T1.2	Is it normal? How do you know? (status + normal range)	normal	☑	☐	☐
T1.3	Should you do anything next? (as the report says)	None	☑	☐	☐
T2.1	Which signals does the study record? (all four)	ATI	☑	☐	☐
T2.2	What does one sample show? (the signal + one finding)	?	☑	☐	☐
T2.3	Can you download the full data now? (no: request + agreement)	request?	☐	☐	☐

Questionnaire 1 = strongly disagree 5 = strongly agree

		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☐	☑
2	I found the system unnecessarily complex.	☐	☑	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☑
4	I think that I would need the support of a technical person to use this system.	☑	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☑
6	I thought there was too much inconsistency in this system.	☑	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☑	☐
8	I found the system very cumbersome to use.	☑	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☑	☐
10	I needed to learn a lot of things before I could get going with this system.	☑	☐	☐	☐	☐
E1	Aura Screen explained my results in words I understood.	☐	☐	☐	☐	☑
E2	The website explained what each sample dataset shows.	☐	☐	☐	☑	☐

Debrief

Most confusing moment: _____

What would you change first? _____

Facilitator: SUS score 88 G3 met: T1 ☐ T2 ☐

Appendix B: Filled inspection & accessibility checklist

Lab 9 — Inspection & Accessibility Checklist

fill once, both of us

MMAC Study Hub and Aura Screen (Lab 8 build) Nongmaithem Hans (2024AIB1011) Himanshu Patel (2024AIB1007)

Date checked: _____ Checked by: Himanshu

How: together, walk through T1 (Aura Screen: sign up → scan → report) and T2 (Study Hub: find data → explain a sample → request access) on the improved build. Mark each Lab 7 issue against Nielsen's 10 heuristics, then do the accessibility checks.

1. Lab 7 issues — fixed or not?

F = fixed, P = partly fixed, N = not fixed. "Not changed yet" = no Lab 8 change; check if it still happens.

ID	Lab 7 problem	Sev	What we changed in Lab 8 (Fig no.)	F	P	N
Aura Screen						
A1	Phone/kiosk view only via dev tools	3	Phone / Kiosk / Full toggle (13)	☑	☐	☐
A2	Scan Hub scrolls in phone view	3	No-scroll phone screens	☑	☐	☐
A3	Third heart-rate tile cut off	3	Shorter, fixed report (13)	☐	☑	☐
A4	Navigation out of thumb reach	2	Bottom tabs: Scan, Report, Timeline (13)	☑	☐	☐
A5	13 px inputs make iOS zoom	2	16 px inputs	☑	☐	☐
A6	Hover stays after a tap	1	Not changed yet	☑	☐	☐
A7	Kiosk screens leave empty space	2	Kiosk-first layout	☑	☐	☐
Study Hub website						
W1	Dashboard links lead nowhere	4	Dashboards built into the site; **EMG and rPPG dashboards added**; dead demo removed (4)	☑	☐	☐
W2	Dashboards are recording tools	3	View-only, 6 widgets each (9, 10)	☑	☐	☐
W3	Dashboards look unlike the site	3	One shared layout for all four dashboards (9, 10)	☑	☐	☐
W4	Site fails to start	3	One-command launcher + README	☑	☐	☐
W5	Placeholder text	2	Placeholder card removed (4)	☑	☐	☐
W6	Open downloads, no terms of use	2	Request access + data use agreement (12)	☑	☐	☐
W7	Charts work with a mouse only	2	Not changed yet	☑	☐	☐
W8	Draggable widgets, no hint	1	Grip icon + "Drag widgets by their title" (9)	☑	☐	☐
W9	Request needs an email app	2	Not changed yet	☑	☐	☐
W10	Thermal values are illustrative	1	"Public sample" badge, sample explained (7)	☑	☐	☐
Found by users in Lab 7						
U1	Report numbers not explained	3	Plain verdict + steps with normal range (13)	☑	☐	☐
U2	Sample data has no explanation	3	*About this data*, caption on every chart, full details (7–12)	☑	☐	☐
U3	Camera blocked, no message	3	Camera errors explained on screen	☑	☐	☐
U4	Hard to find the data	2	Explore button, *Data* menu with descriptions, links, footer (1–3, 6)	☑	☐	☐
U5	Access form long and dense	2	3 short steps + "In short" summary (12)	☐	☑	☐
U6	Google is the only sign-in	1	Email + password added	☐	☑	☐
U7	Unclear how the two products relate	1	*Built with the data* section (4)	☐	☑	☐
Totals			Fixed _____ Partly _____ Not fixed _____ (out of 24)			

New problems found this round (screen — problem — heuristic — severity 0–4):

2. Accessibility check — before vs after

Write Pass, Partly or Fail in the Lab 9 column.

Check	How we check it	Lab 7	Lab 9	What we saw
Text contrast	Faded grey text readable; DevTools contrast ≥ 4.5:1	Fail	Pass	lowest ratio: 5.59
Chart labels	Axis and phase labels readable at a glance (Fig 11)	Fail	Pass	
Text size	Captions, labels and badges big enough	Partly	Pass	
Keyboard	Whole site usable with keyboard only (menu, dashboards, request form, charts)	Partly	Pass	
Labels	Images and icon buttons have names (theme toggle, close)	Partly	Yes	
Zoom / touch	200 % zoom loses nothing; Aura buttons ≥ 44 px	new	No	
Lighthouse	Accessibility score: home ___ /ecg ___ /emg ___ /rppg ___ Aura ___	new	Fail	